

$$M = 10.3 \text{ kg}$$

7 Ans: C

Net force

= Upthrust by air – Weight of helium – Weight of balloon skin & instrument

= Weight of air displaced – $\rho_{\text{He}} Vg - Mg$

= $\rho_{\text{air}} Vg - \rho_{\text{He}} Vg - Mg$

= $(\rho_{\text{air}} - \rho_{\text{He}}) Vg - Mg$

8 Ans: C